

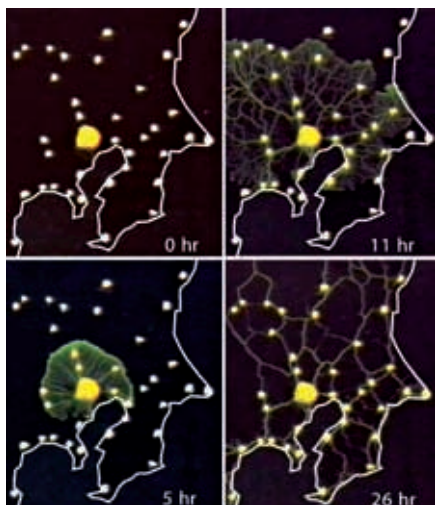
Improve Android

Apps to improve performance of an Android phone
<http://bit.ly/wrnzsT>

A look at slime mould in action

A time lapse vidoe of slime mould solving a mode.
Check it out at
<http://bit.ly/xCGh3w>

Tomorrow's Tech



The Tokyo Rail Network in 24 hours

for transferring the food across the network and amazingly, it can reach speeds of around 1.3 m/s. To put that in perspective, an average person walks as fast as the cytoplasm travels.

So what's so great about that?

What scientists have discovered is that slime mould actually finds the most efficient way to transport food. A fault-tolerant network if you will. The transport system of the mould is filled with redundancies and always

ensures that food will reach the central "hub", even if a strand (or more) is cut.

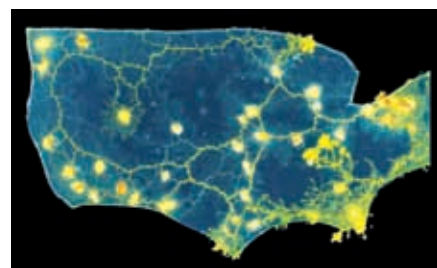
When placed in a simple maze, it was discovered that the slime mould solved it and found the shortest route to the exit, where a food source was placed. This is a small achievement and we're sure even a five year-old can do it, but what about something larger? A rail network for example, or even a complex computer networking problem? These are questions which even a super-computer balks at. The amount of data is too large and very few super-computers can even process this information fully. Even so, super-computing time is very expensive and not everyone can get their hands on it.

In a recent experiment, researchers placed a blob of slime mould in an enclosed petri dish and placed food sources (oat flakes, in this case), at strategic points around the dish. The mould was left to its own devices for about 24 hrs, after which the scientists examined the final results or network that had formed. What they discovered was startling, to say the least. The mould hadn't been placed in just any petri dish, neither was the placement of the food random. The petri dish was shaped like a map of Tokyo and the food was placed at the

railway stations along the Tokyo Rail Network! What startled the researchers was that this single-celled protoplasmic lump had formed a network which coincided almost exactly with this network. This is even more amazing considering that the Tokyo Rail Network is among the most efficient and fault tolerant in the world!

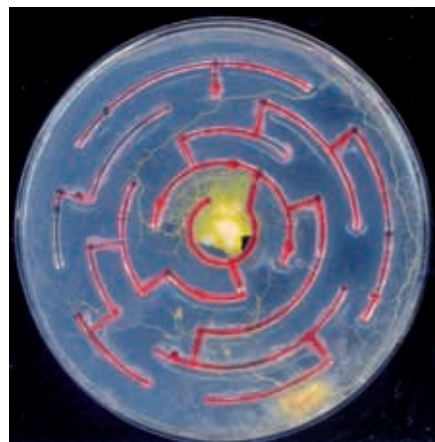
Now what?

Slime mould has proved its mettle against the best super-computers and even humans in the world. What took engineers and computers years to do (refining the Tokyo Rail Network), the slime mould did in 24 hours. This aspect of it has astounded the



US road and power grid in Slime mould

scientists the world over, and the only conclusion they could come to is that efficient networking is one of the most abstruse and complex sciences in the world, but one that nature has mastered over millenia of evolutionary change. Biological organisms like slime mould have provided scientists with a readily scalable solution to growing networks in general. The networks so produced are

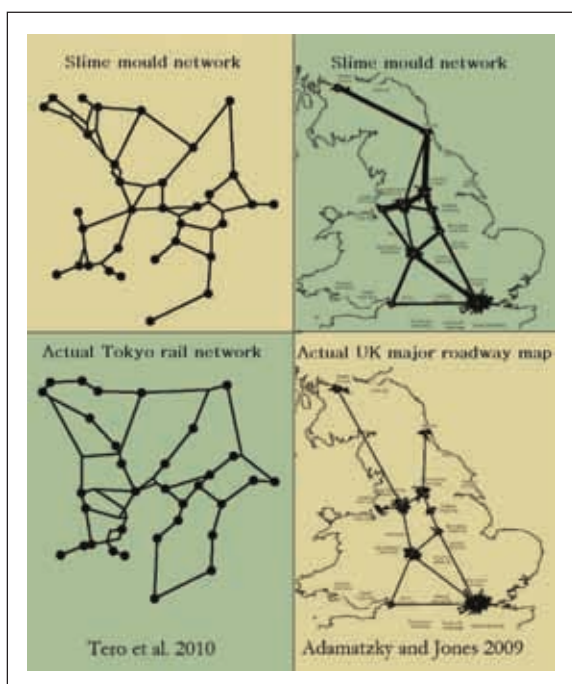


Slime maze

quite robust and fault tolerant as well. Scientists and mathematicians are also working on developing a mathematical model based on this phenomena- something they hope that can be applied to all aspects of network design, including social networking, the laying of power lines, telephone lines and much more.

Even the chipset manufacturing industry can see a revolution with this sort of technology. Imagine what it would be like, a world of slime mould farms busily pulsating away, solving mankind's most acute networking problems!

It is fascinating and in many ways, a humbling experience to see what nature has to teach us. If a single-celled, brainless lump can teach us so much, what else is there waiting to be discovered? ■



Slime mold maps